AP Statistics — Unit 5

Sample Test — Sampling Distributions

Name: ____________________________ Date: ________________ Period: _________

Directions: Show all work for free response. For multiple choice, circle the best answer. Use the Normal distribution tables or your calculator as needed.

Part I — Multiple Choice (2 pts each × 10 = 20 pts)

1. Which of the following is the best description of a *sampling distribution*?
   (A) The distribution of data values in a single sample.
   (B) The distribution of all possible values of a statistic over all possible samples.
   (C) A histogram of the population from which samples are drawn.
   (D) The probability that a sample mean equals the population mean.
   (E) The distribution of sample sizes used in a study.

2. Heights of adult males are approximately $N(69.2 \text{ in}, 3.6 \text{ in})$. What is $P(X > 75)$?
   (A) 0.9463
   (B) 0.0537
   (C) 0.1611
   (D) 0.8389
   (E) 0.0268

3. A population has $\mu = 50$ and $\sigma = 10$. A random sample of $n = 25$ is taken. What is $\sigma_{\bar{x}}$?
   (A) 10
   (B) 5
   (C) 2
   (D) 0.4
   (E) 50

4. The Central Limit Theorem guarantees that the sampling distribution of $\bar{x}$ is approximately normal when which condition is met?
   (A) The population is normally distributed.
   (B) $n \geq 10$.
   (C) $n \geq 30$ (or the population is normal).
   (D) $np \geq 10$ and $n(1-p) \geq 10$.
   (E) The sample size is at least 5% of the population.

5. Which of the following statistics is *not* an unbiased estimator of its corresponding population parameter?
   (A) $\bar{x}$ as an estimator of μ.
   (B) $\hat{p}$ as an estimator of p.
   (C) s^2 (computed with $n-1$) as an estimator of σ^2.
   (D) The sample maximum as an estimator of the population maximum.

(E) The sample proportion as an estimator of the population proportion.

6. In a large city, 40% of residents have a gym membership ($p = 0.40$). A random sample of $n = 100$ residents is selected. The standard deviation of the sampling distribution of $\hat{p}$ is:

   (A) 0.40

   (B) 0.0490

   (C) 0.24

   (D) 0.16

   (E) 0.0024

7. A random sample of $n = 64$ adults is selected from a population with $\mu = 120$ and $\sigma = 16$. Which of the following is the probability $P(\bar{x} > 122)$?

   (A) 0.8413

   (B) 0.1587

   (C) 0.4013

   (D) 0.0548

   (E) 0.3085

8. To use the normal model for the sampling distribution of $\hat{p}$, which condition must be verified?

   (A) $n \geq 30$

   (B) $\sigma > 0$

   (C) $np \geq 10$ and $n(1-p) \geq 10$

   (D) The population is symmetric.

   (E) $p > 0.5$

9. A researcher takes random samples of size $n = 50$ from Group 1 ($\mu_1 = 80$, $\sigma_1 = 12$) and $n = 50$ from Group 2 ($\mu_2 = 75$, $\sigma_2 = 10$). The mean of the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is:

   (A) 155

   (B) 5

   (C) -5

   (D) 7.81

   (E) 2.24

10. Which of the following best explains why a larger sample size produces a more precise estimate of a population parameter?

    (A) Larger samples eliminate bias.

    (B) Larger samples increase $\sigma_{\bar{x}}$.

    (C) Larger samples reduce the standard error, narrowing the sampling distribution.

    (D) Larger samples guarantee a normal population.

    (E) Larger samples make the statistic equal to the parameter.

Part II — Free Response (see point values)

11. **(12 pts)** According to national data, 28% of adults exercise at least 5 days per week ($p = 0.28$). A health researcher surveys a random sample of $n = 200$ adults from a large population.

(a) Check all three conditions for using the normal model for $\hat{p}$. State each condition and show that it is (or is not) satisfied. (3 pts)

(b) Find $\mu_{\hat{p}}$ and $\sigma_{\hat{p}}$. (2 pts)

(c) Find $P(\hat{p} > 0.32)$. Show all steps: standardize, and state the probability. (4 pts)

(d) Interpret $P(\hat{p} > 0.32)$ in a complete sentence in context. (3 pts) ________________

12. **(18 pts)** A manufacturer produces bolts with lengths that are approximately normally distributed with $\mu = 50$ mm and $\sigma = 1.5$ mm.

(a) What is the probability that a single randomly selected bolt has length greater than 51.5 mm? (3 pts)

(b) A quality inspector measures a random sample of $n = 36$ bolts each shift. Check all three conditions for using the normal model for $\bar{x}$. (3 pts)

(c) Find $\mu_{\bar{x}}$ and $\sigma_{\bar{x}}$. (2 pts)

(d) Find $P(\bar{x} > 50.5)$. (4 pts)

(e) Compare your answers to (a) and (d). Explain why the probability for $\bar{x}$ is much smaller than the probability for a single bolt. (3 pts) ____________________

____

____

(f) The manufacturer is concerned if the sample mean falls below 49.4 mm. Find this probability and determine if this should be considered an unusual event. (3 pts)

13. **(10 pts)** A study compares the proportion of students who complete homework in two schools. At School 1, 65% complete homework ($p_1 = 0.65$, $n_1 = 120$). At School 2, 58% complete homework ($p_2 = 0.58$, $n_2 = 150$). The samples are independent random samples from large student populations.

(a) Verify the four Large Counts conditions. (4 pts)

(b) Find the mean and standard deviation of the sampling distribution of $\hat{p}_1 - \hat{p}_2$. (3 pts)

(c) Find $P(\hat{p}_1 - \hat{p}_2 > 0.15)$. (3 pts)

Part I: __________ / 20 **Part II #11:** __________ / 12 **Part II #12:** __________ / 18 **Part II #13:** __________ / 10 **Total:** __________ / 60